

Variable Selection for Kernel Two-Sample Tests

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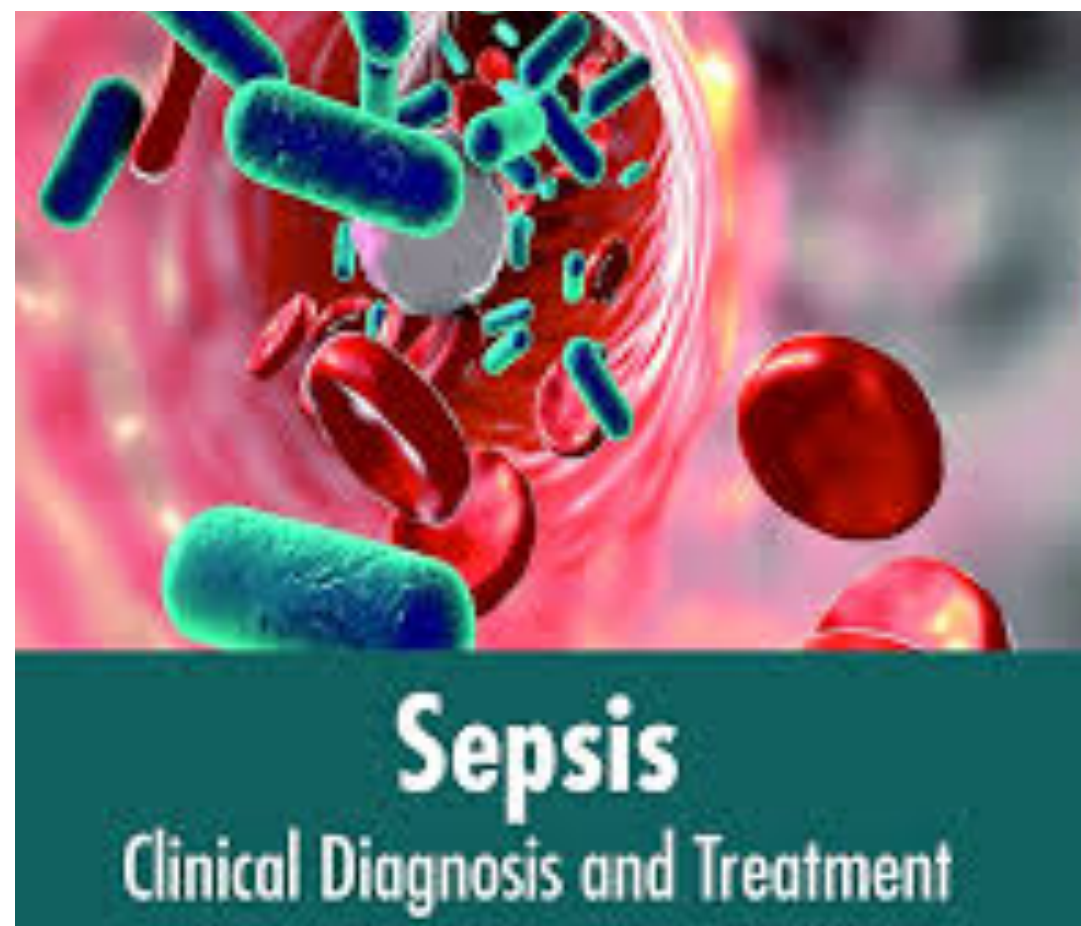
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Problem Setup

- ❖ **Given:** *high-dimensional* samples from unknown distributions μ and ν
- ❖ **Q1:** same distribution or not (**two-sample testing**)?
- ❖ **Q2:** which subset of variables differentiates two sets of samples (**variable selection**)?



Literature on Variable Selection

- Existing methods are designed for classification/regression problems:

$$\min_{w \in \mathbb{R}^D} \sum_{i=1}^n \ell(y_i, w^\top x_i) \quad \text{s.t. } \|w\|_0 \leq d$$

- Solved as mixed-integer program (MIP)
- Convex relaxation, e.g., Lasso
- Our problem is different (features can be **nonlinear**, different **criteria**)

Bertsimas, King, Mazumder, AoS-16

Bertsimas, Pauphilet, Parys, ML-21

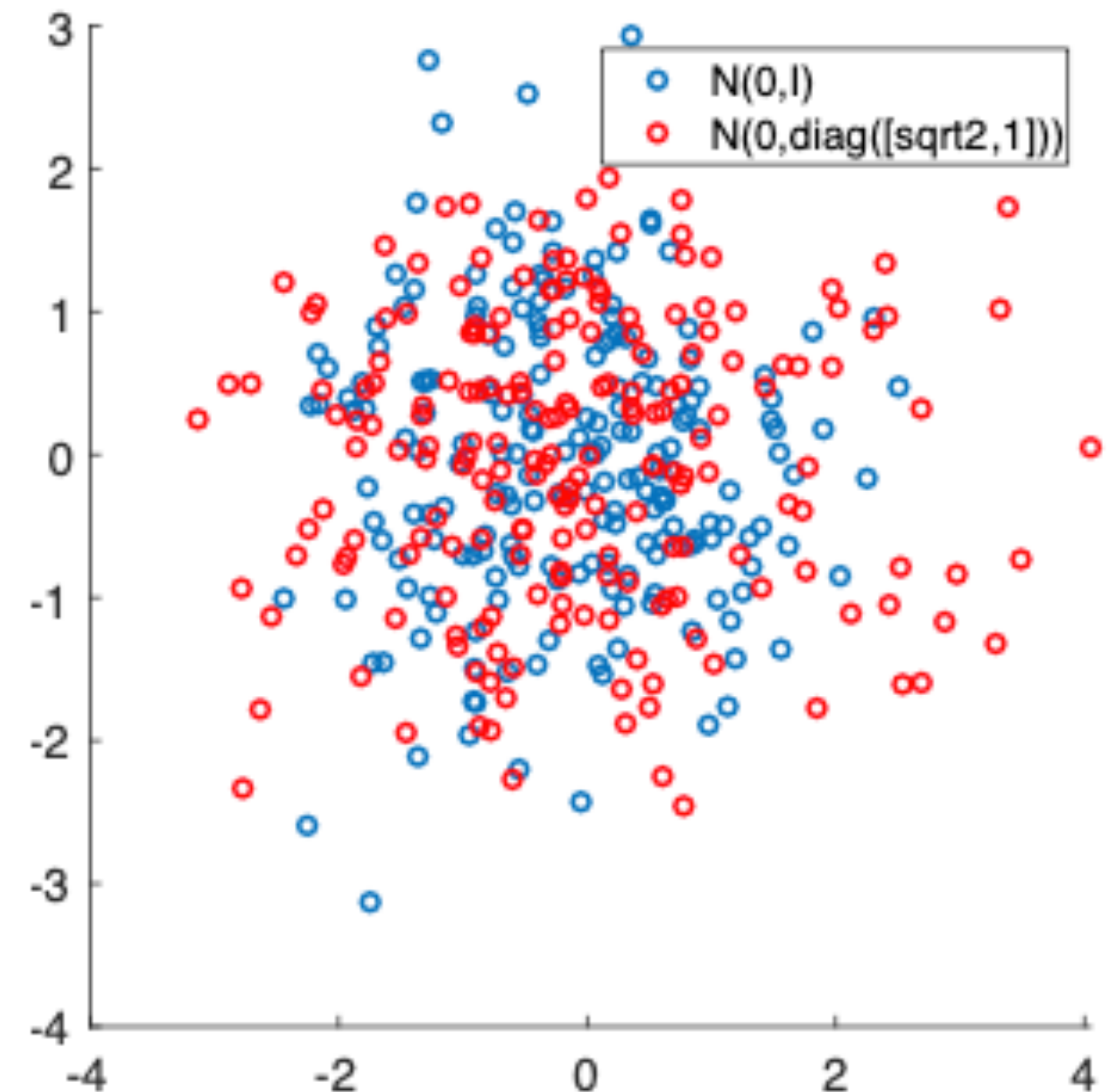
Tibshirani, JRSSB-96

Why sparse logistic regression is not enough?

- Consider $\mu = \mathcal{N}(0, I_D)$, $\nu = \mathcal{N}(0, \text{diag}(\sqrt{2}, 1))$
- Logistic regression produces a linear testing statistic:

$$T := \frac{1}{n} \sum_i w^\top x_i - \frac{1}{n} \sum_i w^\top y_i$$

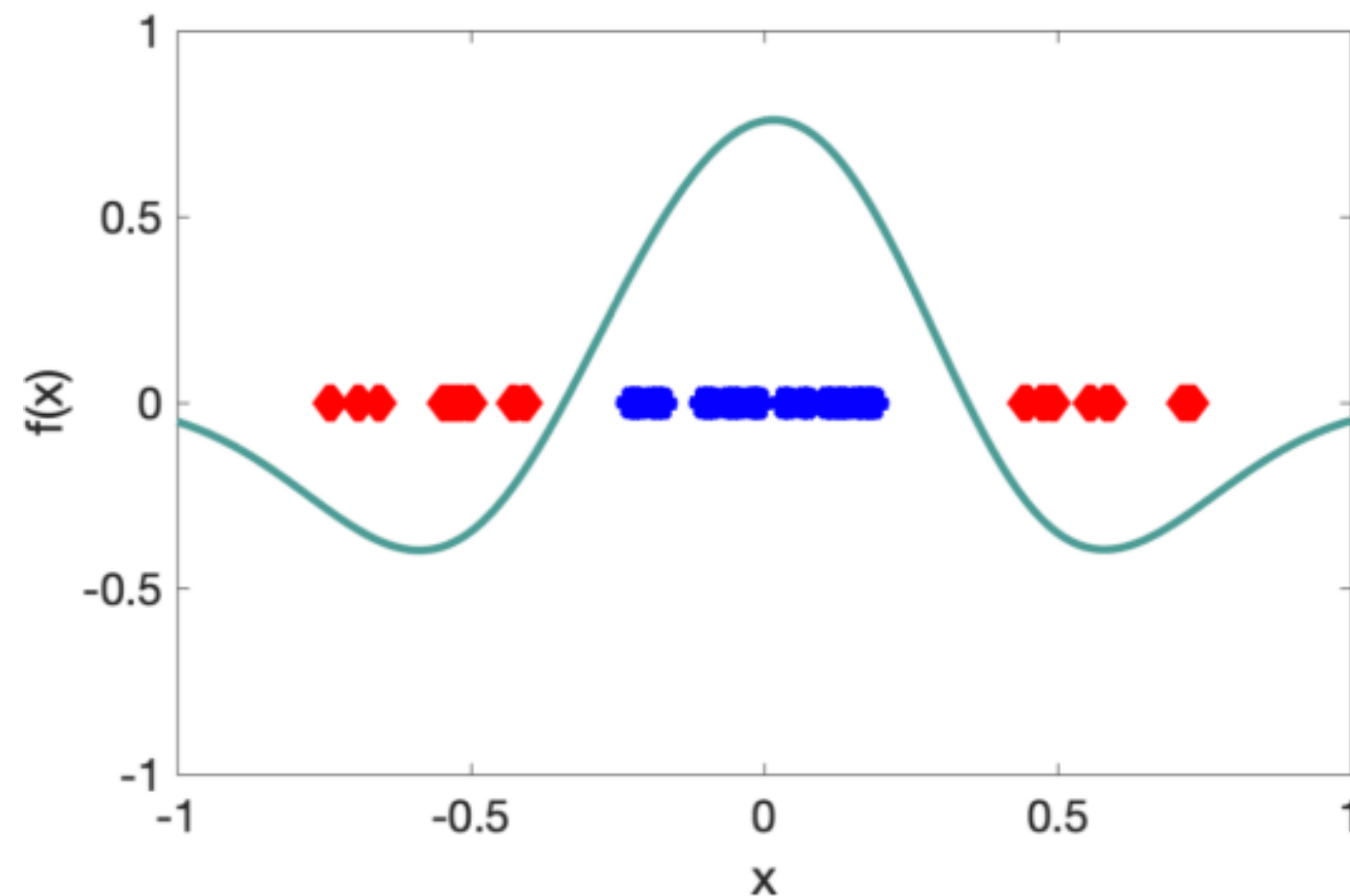
- $\mathbb{E}[T] = 0, \forall w$
- Logistic regression cannot identify difference in the 2nd moment



Kernel Maximum Mean Discrepancy

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) = \sup_{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \leq 1} \mathbb{E}_{x \sim \mathbb{P}}[f(x)] - \mathbb{E}_{y \sim \mathbb{Q}}[f(y)]$$

- $\mathcal{H}$: reproducing kernel Hilbert space (RKHS) dependent on $k(\cdot, \cdot) : \mathbb{R}^{D \times D} \rightarrow \mathbb{R}$
- Optimal test function when $\mathbb{P} \neq \mathbb{Q}$:



Kernel MMD for Two-sample Testing

$$\text{MMD}(X, Y) = \frac{1}{n(n-1)} \sum_{i,j \neq i} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i,j \neq i} k(y_i, y_j) - \frac{2}{n^2} \sum_{i,j} k(x_i, y_j)$$

$k(X, X)$	$k(X, Y)$
$k(X, Y)$	$k(Y, Y)$

- Reject $H_0 : \mu = \nu$ if $\text{MMD}(X, Y) > \Delta$
- **Efficient** for high-dimensional testing
- Not **interpretable**

Variable Selection using MMD

❖ **Given:** kernel $k(x, y) : \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$

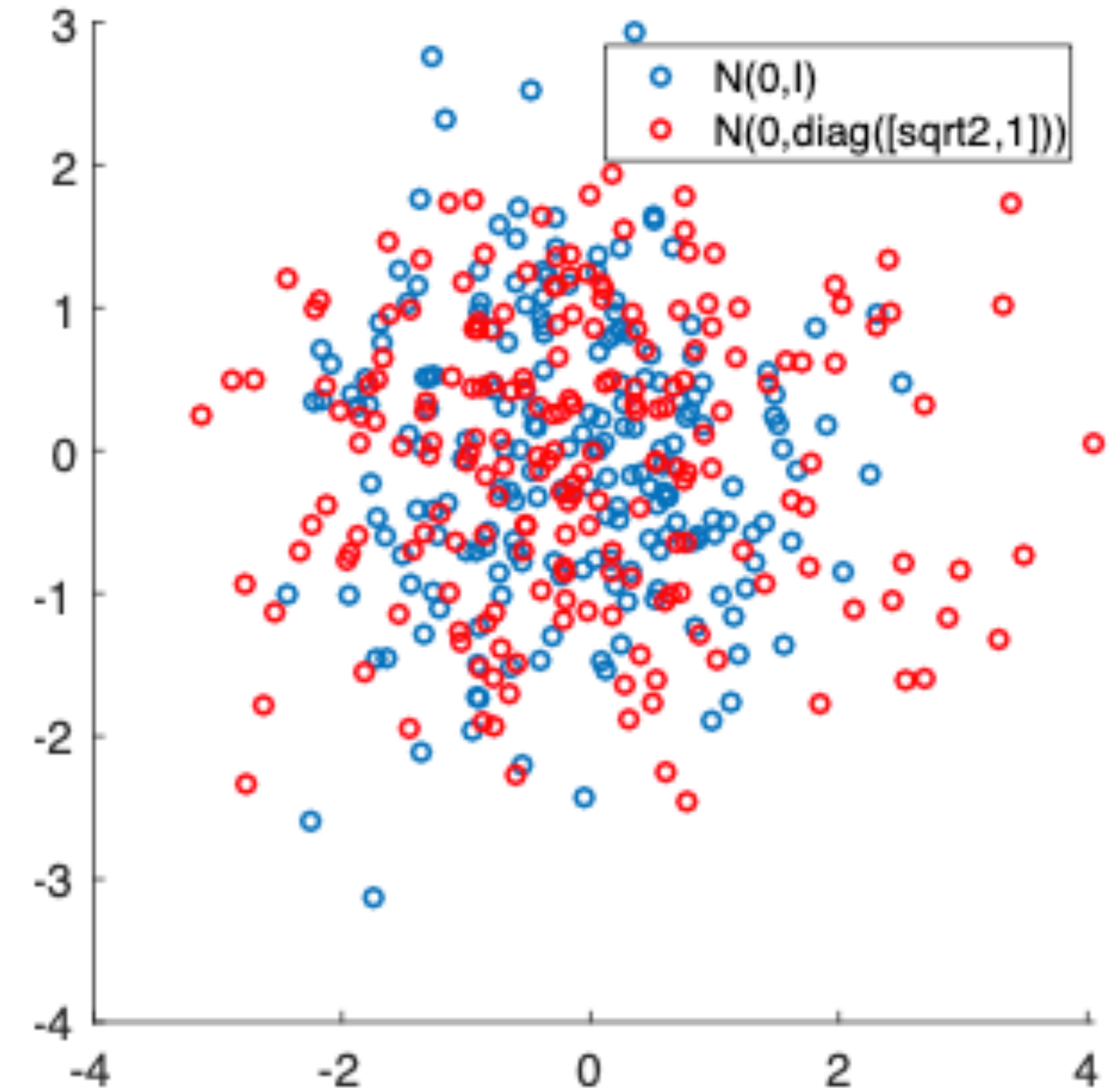
- **Linear:** $k(x, y) = x^\top y$
- **Quadratic:** $k(x, y) = \left(\sum_s x^{(s)} y^{(s)} + c \right)^2$
- **Gaussian:** $k(x, y) = \exp\left(\|x - y\|_2^2 / \sigma^2 \right)$

❖ Selection of variables encoded by vector $z \in \mathbb{R}^D$

$$z^{(s)} = \begin{cases} \text{nonzero}, & \text{if } s \in S, \\ 0, & \text{otherwise} \end{cases}$$

❖ **Criterion:**

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} \text{MMD}(z \odot X, z \odot Y)$$



$$z \odot X = (z^{(1)} X^{(1)}, \dots, z^{(D)} X^{(D)})$$

Linear Kernel Case

$$k(z \odot x, z \odot y) = \sum_s (z^{(s)})^2 x^{(s)} y^{(s)}$$

$$\text{MMD}(X, Y) = \frac{1}{n(n-1)} \sum_{i,j \neq i} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i,j \neq i} k(y_i, y_j) - \frac{2}{n^2} \sum_{i,j} k(x_i, y_j)$$

Linear Kernel Case

$$k(z \odot x, z \odot y) = \sum_s (z^{(s)})^2 x^{(s)} y^{(s)}$$

❖ **Reformulation:**

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_1 = 1} a^\top z$$

$$a^{(s)} = \left(\frac{1}{n} \sum_i x_i^{(s)} - \frac{1}{n} \sum_i y_i^{(s)} \right)^2$$

❖ **Pros:** simple solution available

❖ **Cons:** limited testing power (discriminative power)

❖ **Extension:** $k(z \odot x, z \odot y) = \sum_s (z^{(s)})^2 k'(x^{(s)}, y^{(s)})$

Quadratic Kernel Case

$$k(z \odot x, z \odot y) = z^\top A(x, y)z + z^\top t(x, y) + \text{constant}$$

$A(x, y) \in \mathbb{R}^{D \times D}$, $t(x, y) \in \mathbb{R}^D$ are input parameters dependent on (x, y)

❖ **Reformulation of quadratic kernel MMD:**

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$

❖ **Sparse Trust Region Subproblem (STRS)**, an extended version of (**Sparse PCA**)

❖ **Gaussian kernel:** no simple exact algorithm, solved by sequential convex approximation heuristic

Bertsimas D, Cory-Wright R, Pauphilet J, JMLR2022

Li Y, Xie W. IJOC2024

Sparse Trust Region Subproblem

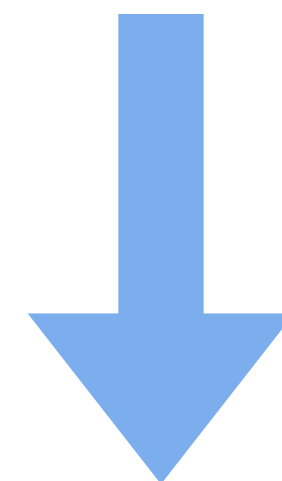
$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$

- Nonconvex Mixed-Integer program in general
- Reformulation needed to develop tractable algorithms

Sparse Trust Region Subproblem

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$

- Exact mixed-integer SDP (MISDP) approach:



$$\tilde{A} = \begin{bmatrix} 0 & \frac{1}{2}t^\top \\ \frac{1}{2}t & A \end{bmatrix}, \quad Z = \begin{bmatrix} 1 \\ z \end{bmatrix} \begin{bmatrix} 1 \\ z \end{bmatrix}^\top$$

$$\max_{Z \in \mathcal{S}_{D+1,q}^+} \langle \tilde{A}, Z \rangle$$

$$\text{s.t. } Z^{(i,i)} \leq q^{(i)}, i \in [D],$$

$$Z^{(0,0)} = 1, \text{Tr}(Z) = 2,$$

$$q \in \mathbb{F}^D, \sum_{i \in [D]} q^{(i)} \leq d$$

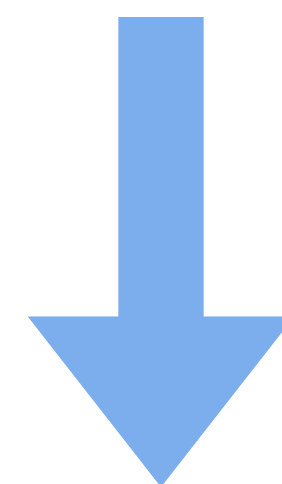
$$\sum_{j \in [D]} (Z^{(i,j)})^2 \leq Z^{(i,i)} q^{(i)}, \quad \left(\sum_{j \in [D]} |Z^{(i,j)}|^2 \right) \leq d Z^{(i,i)} q^{(i)}, i \in [D]$$

**Solved in small-scale with
Benders decomposition**

Approximation Algorithm

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$

- Convex Relaxation approach:



$$\tilde{A} = \begin{bmatrix} 0 & \frac{1}{2}t^\top \\ \frac{1}{2}t & A \end{bmatrix}, \quad Z = \begin{bmatrix} 1 \\ z \end{bmatrix} \begin{bmatrix} 1 \\ z \end{bmatrix}^\top$$

$$\begin{aligned} \max_{Z \in \mathcal{S}_{D+1,q}^+} \quad & \langle \tilde{A}, Z \rangle \\ \text{s.t.} \quad & Z^{(i,i)} \leq q^{(i)}, i \in [D], \\ & Z^{(0,0)} = 1, \text{Tr}(Z) = 2, \\ & q \in [0,1]^D, \sum_{i \in [D]} q^{(i)} \leq d \end{aligned}$$

$$\sum_{j \in [D]} (Z^{(i,j)})^2 \leq Z^{(i,i)} q^{(i)}, \quad \left(\sum_{j \in [D]} |Z^{(i,j)}|^2 \right) \leq d Z^{(i,i)} q^{(i)}, i \in [D]$$

Approximation Algorithm

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$

- Theoretical Guarantee:

$$\text{Optval}(\text{STRS}) \leq \text{Optval}(\text{CVX}) \leq \|t\|_2 + \min \left\{ D/d \cdot \text{Optval}(\text{STRS}), d \cdot \text{Optval}(\text{STRS}) - \min_k |t^{(k)}| \right\}$$

Truncation Algorithms

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$

- **Normalized Sparse Truncation:** Given $z \in \mathbb{R}^D$ and $d \in \mathbb{N}$, define its **sparse truncation** as $\tilde{z}$ that keeps the d largest absolute entries of z only; define $\tilde{z}/\|\tilde{z}\|_2$ as its **normalized sparse truncation**.
- **Truncation Algorithm I:** Solve trust region problem
 - $z^* = \operatorname{argmax}_{z: \|z\|_2=1} z^\top A z + z^\top t$
 - Return $\hat{z}$ as the normalized sparse truncation of z^*
- $\operatorname{Optval}(\text{STRS}) \geq \operatorname{Optval}(\text{Truncation-I}) \geq \frac{d}{D} \operatorname{Optval}(\text{STRS}) - \frac{d}{D} \|t\|_2 - (1 + \sqrt{d/D}) \|t\|_{(d)}$

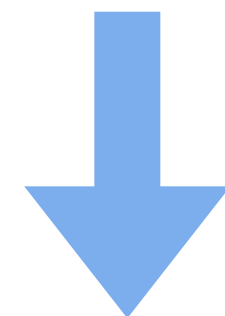
Truncation Algorithms

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$

- **Truncation Algorithm II:**
 - Pick $\hat{z}_i$ as the normalized sparse truncation of $A^{(:,i)}$
 - Return the estimated solution as the best over $\hat{z}_i$'s and e_i 's
- $\text{Optval}(\text{STRS}) \geq \text{Optval}(\text{Truncation-I}) \geq 1/\sqrt{d} \cdot \text{Optval}(\text{STRS}) - 2\|t\|_{(d+1)}$

Sparse Trust Region Subproblem

$$\max_{z \in \mathbb{R}^D, \|z\|_0 \leq d, \|z\|_2 = 1} z^\top A z + z^\top t$$



$$\|z\|_2 \leq 1, \|z\|_0 \leq d \implies \|z\|_1 \leq \sqrt{d}$$

- ℓ_1 -relaxation:

$$\begin{aligned} \max \quad & z^\top A z + z^\top t \\ \text{s.t.} \quad & \|z\|_2 \leq 1, \|z\|_1 \leq \sqrt{d} \end{aligned}$$

- Further approximate it using N -node **piecewise linear approximation** (solved with convex integer programming)
- $\text{Optval}(\text{STRS}) \leq \text{Optval}(\text{Relaxation}) \leq 4 \cdot \text{Optval}(\text{STRS}) + \mathcal{O}(1/N^2)$
- Other two polynomial-time approximation algorithms, but the approximation ratio suffers from curse of dimension ($\min(D/d, \sqrt{d})$)

Testing Power

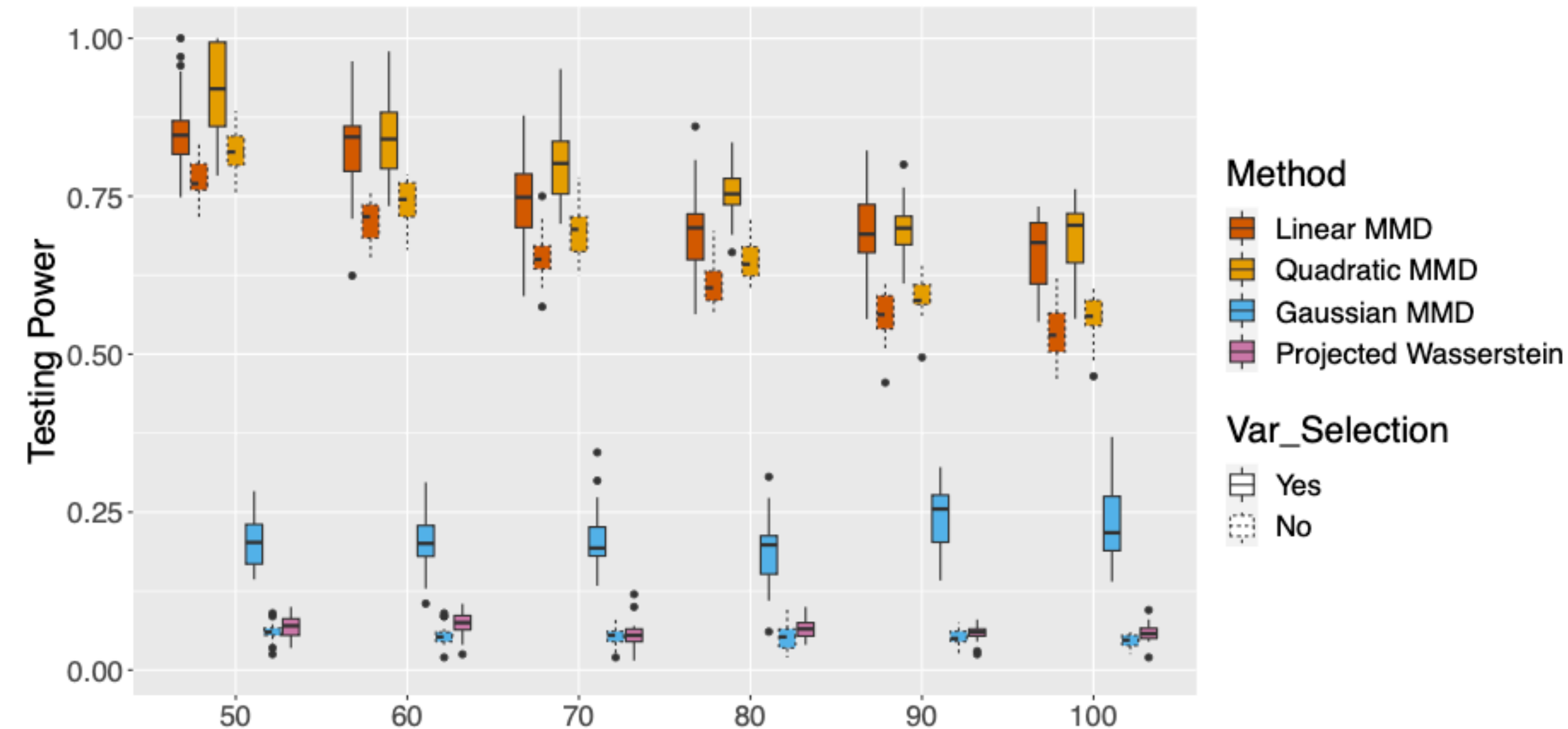
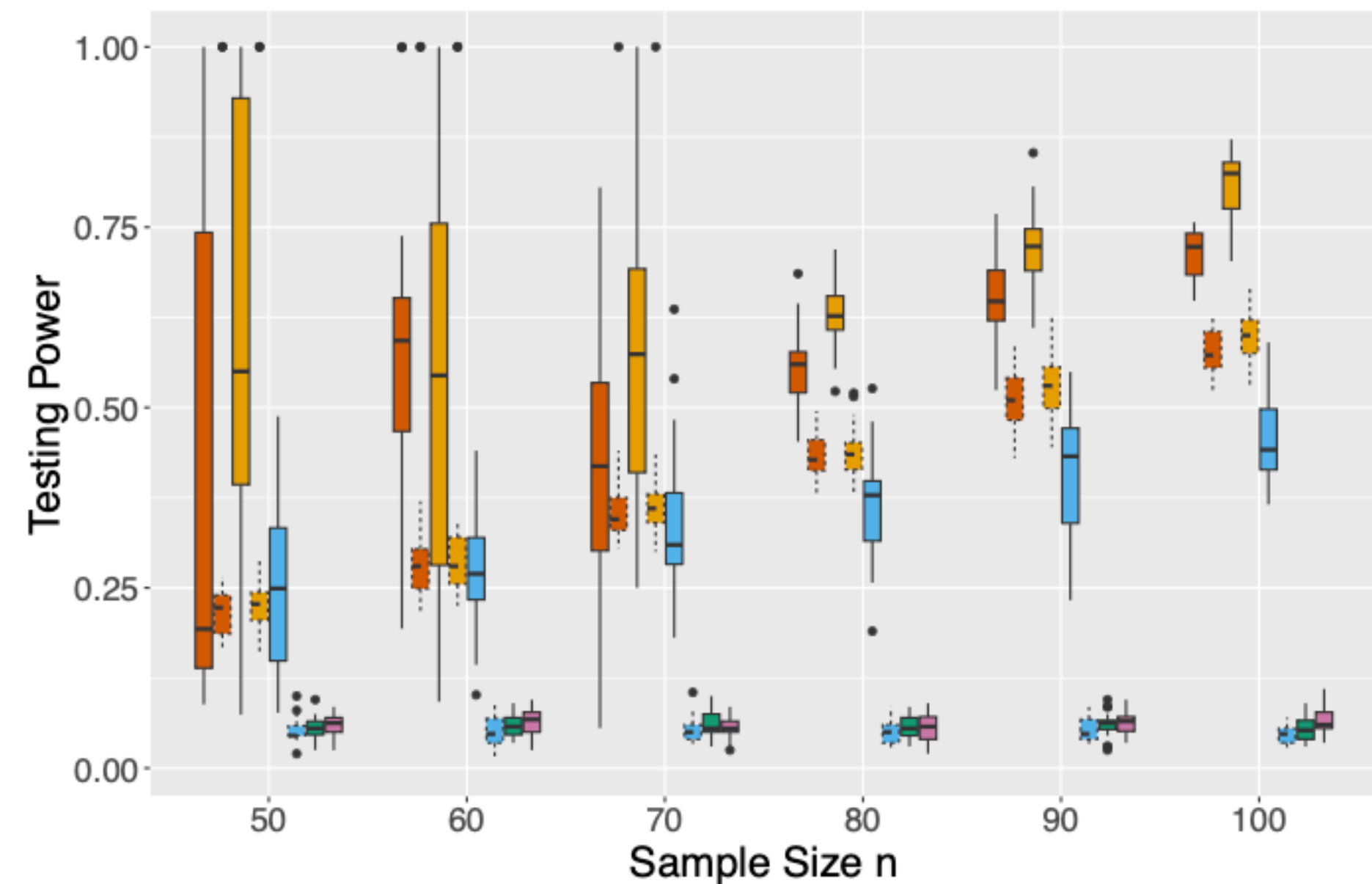
Assumption (Structure of Data):

- **Linear kernel:** There exists s -th coordinate at which $\mu[s] \neq \nu[s]$
- **Quadratic kernel:** There exists (s_1, s_2) such that $\text{Cov}(\mu)_{[s_1, s_2]} \neq \text{Cov}(\nu)_{[s_1, s_2]}$
- **Gaussian kernel:** $\mu \neq \nu$

Theorem (Informal). Fix level $\alpha \in (0, 1/2)$ and specify threshold $\Delta(n, \alpha)$ according to the limiting distribution of testing statistic. Assume MMD optimization is solved exactly. Then:

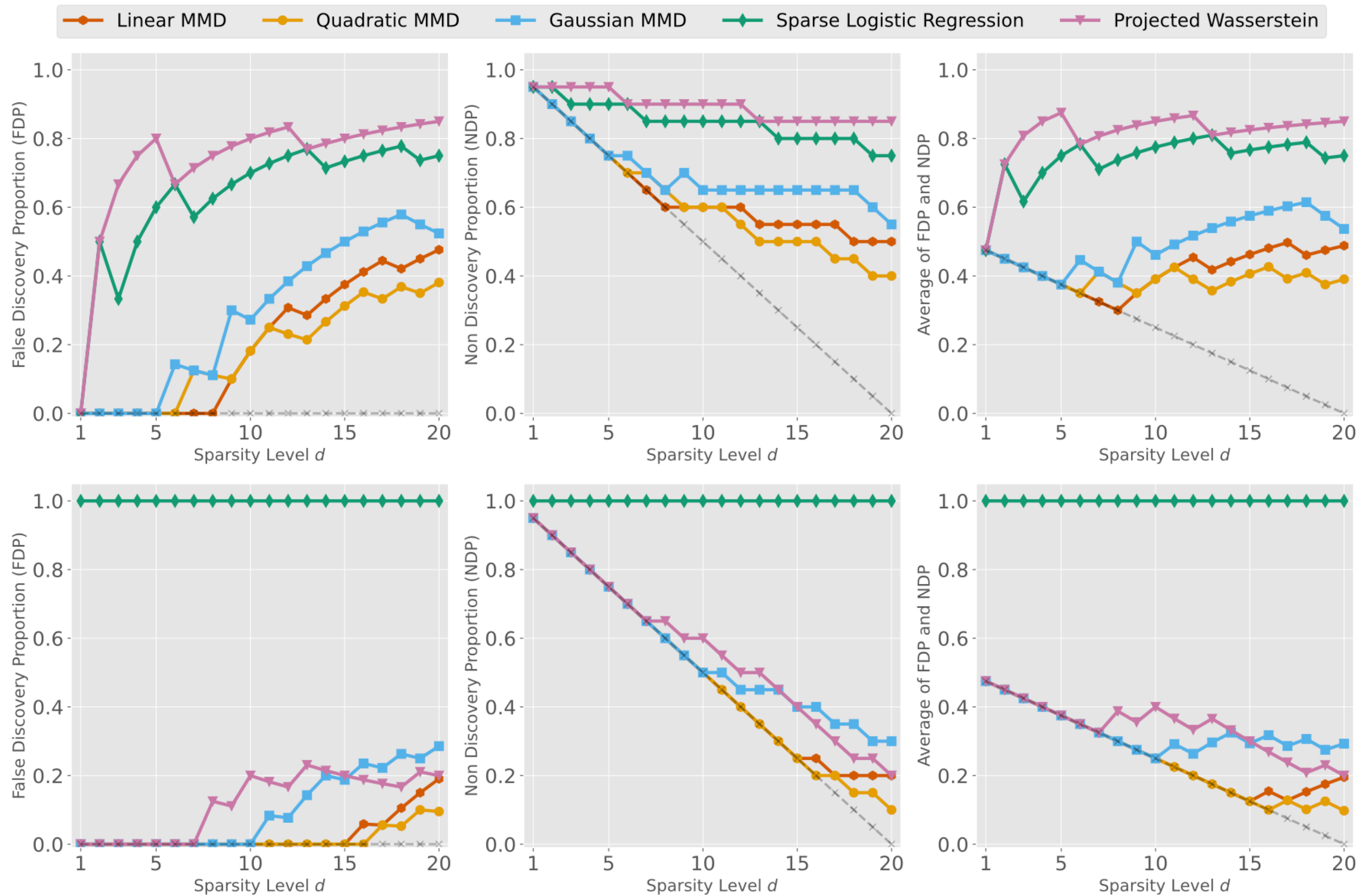
- **Type-I Error** of KMS test is at most α ;
- Under $H_1 : \mu \neq \nu$, the **power** of KMS test is at least $1 - O(n_{Te}^{-1/2})$.

Testing Power v.s. Sample Size/Dimension



- Algorithm complexity: Linear < Quadratic < Gaussian
- Two-sample test performance: Quadratic > Linear > Gaussian

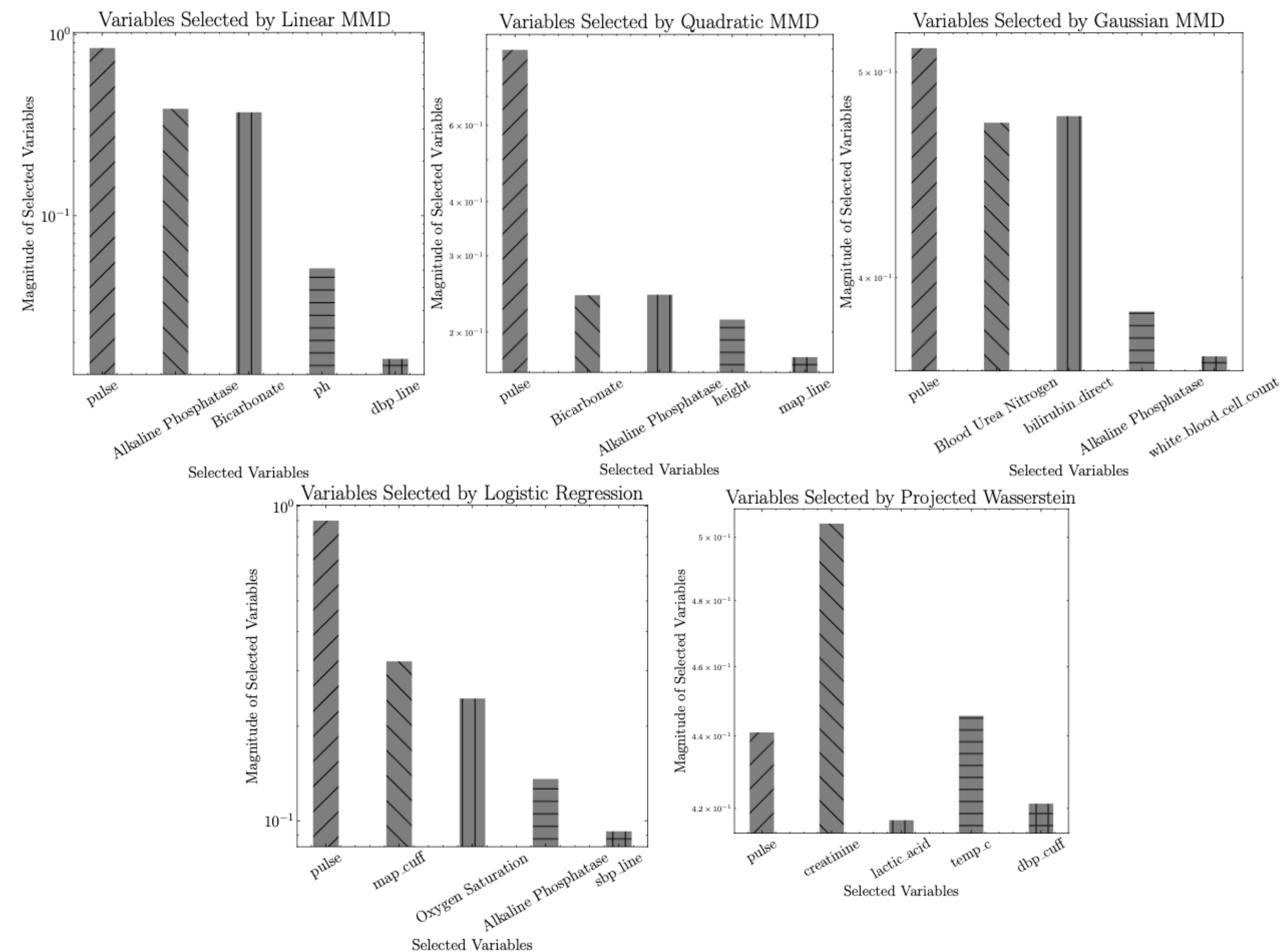
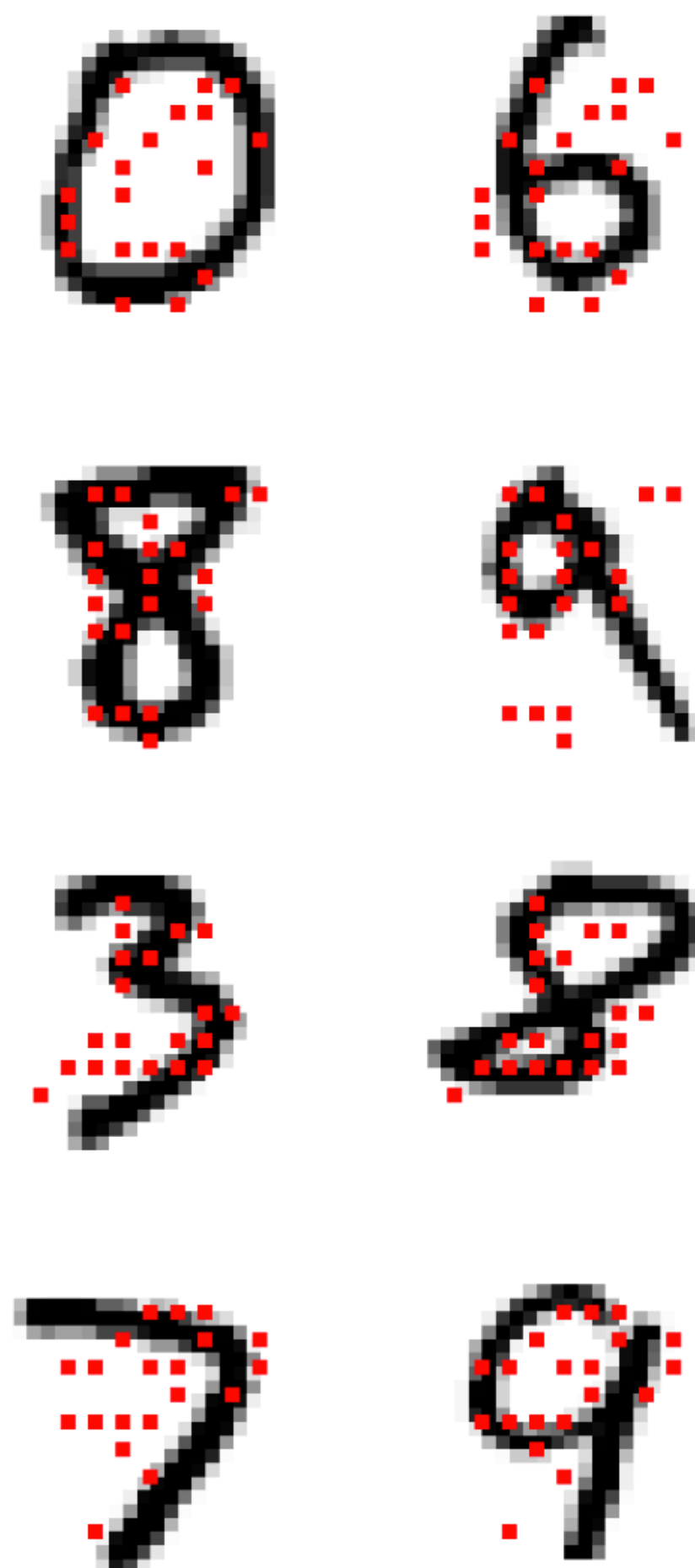
Variable Selection



Interpretability (Real-data examples)

MNIST handwritten digits

Sepsis prediction: $D = 39$, healthy people
 $n_1 = 2077$, patients $n_2 = 2891$



Conclusion

- Provides a variable selection for hypothesis testing framework
- Allows nonlinear feature learning by kernel embedding
- Solves mixed-integer nonconvex program
- Statistical and computational guarantees
- Validate using synthetic and real datasets